regionchecker — Laporan Flow Klasifikasi

Versi 0.2.0 (P10–P12) · 2026-04-18

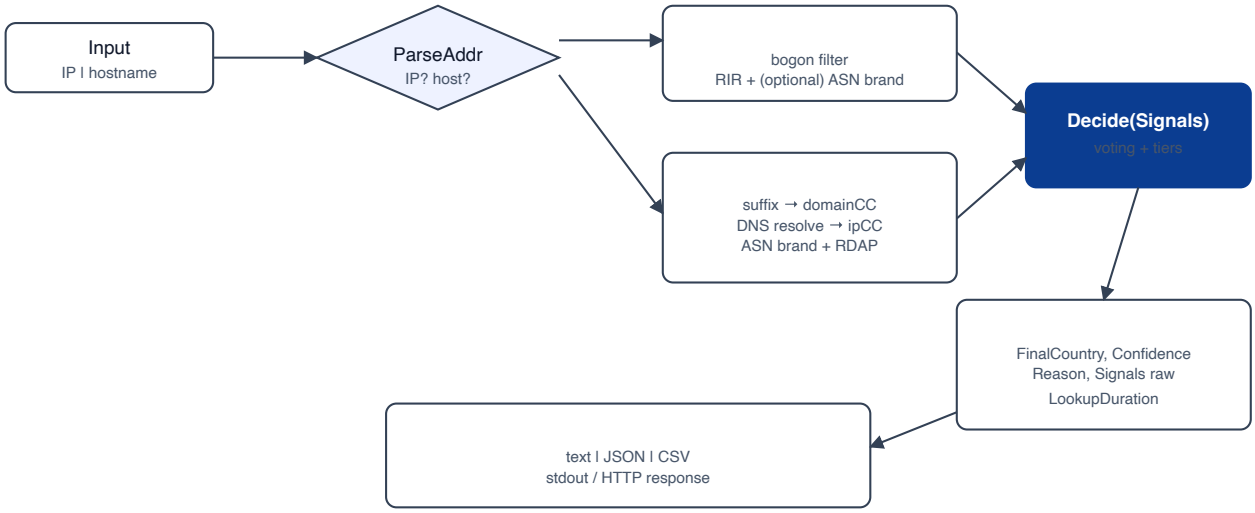
Dokumen: alur lengkap sinyal dari input IP/domain sampai `FinalCountry`. Audiens: developer / reviewer.

1. Gambaran Besar

regionchecker memutuskan **country code** suatu IP atau domain berdasarkan gabungan **empat sinyal**. Tiap sinyal independen dan opsional. Sinyal yang tersedia diberi suara; mayoritas (2+) menang dengan confidence `high`. Bila tidak ada mayoritas, tiered fallback menentukan hasil.

Sinyal	Sumber	Biaya	Aktivasi
Domain suffix	ccTLD list / IDN / geographic gTLD / PSL (offline)	ns	selalu (domain only)
IP geo	RIR delegated files (NRO/APNIC/ARIN/...)	~20ns	selalu
ASN brand	MMDB org name → regex match	µs	`--mmdb` / `--all`
RDAP registrant	IANA RDAP bootstrap → registry → registrar chain (online)	~500–2000ms	`--rdap` / `--all`

2. High-Level Pipeline



3. Branch Detail

3.1 IP Branch (`classifyIP`)

1. `netip.ParseAddr` → `addr.Unmap()` (IPv4-mapped IPv6 diturunkan ke IPv4).

2. `bogon.Match` : kalau reserved (RFC1918, CGNAT, loopback, link-local, multicast, doc) → return `ErrBogon` . Tanpa lookup lebih jauh.
3. `rir.DB.LookupIP` (mmap snapshot, binary search) → `(cc, meta)` . Alokasi per RIR (APNIC/ARIN/...), bukan routing.
4. Opsional `ASN.Lookup(addr)` (MMDB) → `(asn, org)` , lalu `asn.BrandCountry(org)` → `asnCC` .
5. `Decide(Signals{IPCC, ASNCC, IsIPInput: true})` .

> **Catatan:** RDAP tidak dipanggil untuk raw IP (RDAP domain endpoint, bukan IP).

3.2 Host Branch (`classifyHost`)

1. `domain.Country(input)` : pakai ccTLD list (248 entry dari IANA), IDN Punycode, geographic gTLD, PSL. Menghasilkan `domCC` + `domType` (`cctld` / `idn` / `geo-gtld` / `generic`).
2. `Resolver.Resolve(ctx, host)` → list of `netip.Addr` . Timeout default 5s (flag `--timeout`).
3. Pick first non-bogon address → `firstPublic` .
4. `rir.DB.LookupIP(firstPublic)` → `(ipCC, meta)` .
5. Opsional `ASN.Lookup(firstPublic)` + `BrandCountry` → `asnCC` .
6. Opsional `RDAP.Lookup(ctx, input)` → `rdapCC` .
7. Kalau `dnsFailed` dan semua sinyal kosong → `ErrUnresolvable` .
8. `Decide(Signals{...semua sinyal...})` .

4. Decide — Voting & Tiers

4.1 Algoritma

```
Signals = { DomainCC, DomainType, IPCC, ASNCC, RDAPCC, DNSFailed, IsIPInput }

1. IF IsIPInput AND ASNCC=="" AND RDAPCC="":
     IF IPCC != "" → (IPCC, ip-only)
     ELSE          → ("", unknown)

2. votes[cc] = count of non-empty signals == cc
   best = argmax(votes);  n = votes[best]
   IF n >= 2 → (best, high, "X+Y agree on <CC>")

3. IF ASNCC != "" → (ASNCC, high-asn-brand)
4. IF RDAPCC != "" → (RDAPCC, high-rdap-registrant)

5. IF DNSFailed:
     IF DomainCC != "" → (DomainCC, low-dns-failed)
     ELSE              → ("", unknown)

6. IF DomainCC != "" AND IPCC != "" AND DomainCC != IPCC:
     IF DomainCC == "ID" → (ID, medium-domain-id-offshore-host)
     ELSE                → (DomainCC, medium-domain-cc-mismatch)

7. IF DomainCC=="" AND (DomainType=="generic" OR "") AND IPCC=="ID":
     → (ID, medium-generic-tld-id-host)

8. Single-signal fallback: DomainCC only / IPCC only / unknown.
```

4.2 Confidence Tiers

Tier	Arti	Contoh
high	2+ sinyal setuju	google.co.id → domain=ID, ip=ID
high-asn-brand	ASN brand override (company identity > hosting geo)	raw IP punya ASN TOKOPEDIA → ID
high-rdap-registrant	Registrant domain kontradiksi IP	tokopedia.com → RDAP=ID, IP=US
medium-domain-id-offshore-host	.id tapi host luar	contoh.id → IP US
medium-generic-tld-id-host	generic TLD tapi host di ID	contoh.com → IP ID
medium-domain-cc-mismatch	cctld ≠ IP (non-ID)	bbc.co.uk → IP US
low-dns-failed	DNS gagal, pakai domain saja	offline+.id host unresolvable
ip-only	Cuma sinyal IP geo	raw IP 8.8.8.8 → US

unknown	Tidak ada sinyal	—

5. Sinyal — Mekanisme Detil

5.1 Domain Suffix

Ordered check: IDN Punycode → ccTLD exact (e.g. `.id` → ID) → geographic gTLD (`.london` → GB) → PSL untuk multi-part suffix (`.co.uk` → GB). Kalau tidak match apapun dan masih gTLD generic → `domainType=generic`, `domainCC=""`.

5.2 IP Geo (RIR)

Build snapshot dari file delegated-stats (NRO combined by default). Tiap entry: `prefix → (cc, registry)`. Decomposed ke CIDR, disimpan sorted array, binary search O(log n). Mmap-backed untuk zero-copy. Lookup ~19ns/op.

> **Gotcha:** RIR = alokasi, bukan routing. 1.1.1.1 = APNIC/AU meski anycast ke seluruh dunia.

5.3 ASN Brand

MMDB reader dukung dua format:

- **MaxMind / DB-IP Lite**: field `autonomous_system_number` (uint) + `autonomous_system_organization` (string).
- **ipinfo**: field `asn` (string "AS12345") + `as_name`.

Setelah `org` didapat, cocokkan regex (word-boundary anchored):

```
ID carriers/ISPs:
  \b(TELKOM|TELKOMSEL|BIZNET|INDIHOME|LINKNET|CBN|INDOSAT|
     AXIATA|SMARTFREN|FIRST[-\s]?MEDIA|HYPERNET|NIAGAHOSTER|
     DEWAWEB|IDWEBHOST|DATACIPTA|MYREPUBLIC INDONESIA|MORATELINDO)\b

ID internet co. (own ASN):
  \b(TOKOPEDIA|BUKALAPAK|GOJEK|TRAVELOKA|BLIBLI|HALODOC|
     JNE|DETIK|KOMPAS[-\s]?GRAMEDIA)\b
```

> **Perlu ga brand regex?** Kelebihannya: offline, zero latency, jalan bareng IP geo lookup. Keterbatasannya: *hanya hit kalau target pakai ASN sendiri*. Kenyataannya sebagian besar domain host di cloud (AWS/Alibaba/Fastly/Cloudflare) — ASN milik cloud provider, bukan company. Overlap dengan RDAP: RDAP catch "PT Tokopedia" via registrant country apapun hostingnya. **Saran**: pertahankan brand untuk kasus di-ASN-sendiri (cepat, cost nol); tapi jangan andalkan itu doang — RDAP adalah sumber utama untuk identifikasi origin.

5.4 RDAP Registrant

1. Load IANA bootstrap `dns.json` (embed, ~69KB) → map TLD → base URL RDAP registry.
2. `GET <base>/domain/<name>` dengan `Accept: application/rdap+json`, timeout 3s.
3. Parse entities jCard — cari role `registrant`, ambil `adr` parameter `cc` (ISO alpha-2).

4. **Fallback chain**: registry gTLD seperti Verisign .com *tidak* expose registrant, cuma registrar. Ikutin `links[].rel=related, type=application/rdap+json` → fetch registrar RDAP → parse lagi.
5. Disk cache `~/.cache/regionchecker/rdap/<sha256(domain)>.json`, TTL 7 hari. Negative hasil (redacted) juga di-cache untuk hindari re-fetch.

> **Redaksi GDPR/ICANN:** banyak TLD menyembunyikan registrant (especially gTLDs generic). Kalau redacted → `rdapCC=""` → sinyal tidak menyumbang voting. Graceful degrade ke signal lain.

6. Contoh Real

6.1 tokopedia.com (cloud-hosted ID brand)

```
$ ./bin/regionchecker check --all -o json tokopedia.com

{
  "input": "tokopedia.com",
  "resolved": ["47.74.244.18"],
  "domain_suffix": "generic",
  "ip_country": "US",
  "asn": 45102,
  "asn_org": "Alibaba (US) Technology Co., Ltd.",
  "registrant_country": "ID",
  "registry": "arin",
  "final_country": "ID",
  "confidence": "high-rdap-registrant",
  "reason": "rdap registrant ID overrides ip US"
}
```

Flow: DNS → Alibaba US IP. RIR → US. MMDB org = Alibaba → bukan brand ID, `asnCC=""`. RDAP Verisign thin → follow ke CSC Global → registrant PT TOKOPEDIA, cc=ID. Voting: [US, ID] = tidak ada mayoritas. Tier 4 (RDAP sole signal) → `high-rdap-registrant`, FinalCountry=ID.

6.2 google.co.id (domain + host agree)

```
  "domain_country": "ID", "domain_suffix": "cctld",
  "ip_country": "AU", "asn_org": "Google LLC",
  "registrant_country": "ID",
  "final_country": "ID", "confidence": "high",
  "reason": "domain cctld+rdap agree on ID"
```

Voting: ID (domain) + ID (rdap) = 2 votes → `high`. IP AU (Google anycast) diabaikan karena kalah voting.

6.3 8.8.8.8 (raw IP)

```
  "type": "ip", "ip_country": "US",
  "asn_org": "Google LLC",
  "final_country": "US", "confidence": "ip-only",
  "reason": "raw ip input; rir lookup"
```

6.4 bukalapak.com

```
  "ip_country": "US", "asn_org": "Fastly, Inc.",
  "registrant_country": "ID",
  "final_country": "ID", "confidence": "high-rdap-registrant"
```

Fastly CDN di US. ASN brand tidak match. RDAP menyelamatkan.

6.5 traveloka.com (registrant di UK)

```
"ip_country": "US", "asn_org": "Amazon.com, Inc.",
"registrant_country": "GB",
"final_country": "GB", "confidence": "high-rdap-registrant"
```

> RDAP kata GB — memang tuples "Amazon Registrar" listing address UK/privacy proxy. Exact per data, walau operasional Traveloka di Indonesia. Interpretasi hasil: registrant ≠ HQ.

7. CLI Flag Reference

Flag	Efek	Biaya
`--offline`	Skip DNS, domain signal saja	0
`--mmdb <path>`	Aktifkan ASN lookup + brand heuristic	µs/lookup
`--rdap`	Aktifkan RDAP registrant (cache 7d)	~500–2000ms cold
`--rdap-timeout <d>`	Timeout per RDAP call	—
`--all`	Auto-MMDB + RDAP enabled	semua di atas
`--timeout <d>`	Timeout DNS	—
`-o text|json|csv`	Format output	—

Auto-MMDB resolution order (`--all`)

1. `$REGIONCHECKER_MMDB`
2. config `mmdb_path` (yaml)
3. `$XDG_CACHE_HOME/regionchecker/asn.mmdb`
4. `~/.cache/regionchecker/asn.mmdb`
5. `~/.cache/regionchecker/country_asn.mmdb`
6. `~/.local/share/regionchecker/asn.mmdb`
7. `/usr/share/GeoIP/GeoLite2-ASN.mmdb`

8. Evaluasi: Perlu ASN Brand Regex?

Pro (simpan)

- Offline — jalan tanpa network.
- Latency nol extra; sudah dipanggil bareng ASN lookup.
- Catch kasus raw IP owned-ASN (mis. `103.10.169.1` → ASN TOKOPEDIA).

Kontra (drop)

- Daftar hardcoded — perlu maintenance.
- Jarang hit: target umumnya host di cloud (Fastly/AWS/Alibaba/Cloudflare).
- Overlap dengan RDAP registrant (lebih authoritative).

- Confidence tinggi (`high-asn-brand`) kalau hit — company identity stronger than hosting geo.
- Bias regex false positive (e.g. brand name bergemilang di org string lain).

> **Rekomendasi:** pertahankan tetapi **kecilkan cakupan** ke brand yang benar-benar jalanin ASN sendiri (TOKOPEDIA, BUKALAPAK, GOJEK) + ISP besar (TELKOM, BIZNET, INDOSAT). Jangan tambah brand cloud-hosted (TRAVELOKA/BLIBLI/DETIK — tidak jalanin ASN sendiri); mereka akan selalu false-negative di sini. Untuk mereka, RDAP yang pegang.

9. Ringkasan Keputusan Arsitektur

- **Struct Signals** bukan positional args — growth-friendly, readable.
- **Voting** → majority 2+ wins; single-signal fallback tetap ada (backward compat).
- **RDAP chain** — registry thin, follow "related" ke registrar. Parse defensive (tolerate redaction).
- **Disk cache RDAP** — TTL 7d. Negative (redacted) juga di-cache (hindari repeat loss).
- **MMDB dual-format** — MaxMind + ipinfo. Decoder silently ignore absent fields.
- **Graceful degrade** — tiap sinyal opsional, error tidak memblock klasifikasi.

10. File yang Disentuh (P10–P12)

```
internal/asn/orgregex.go          # brand regex diperluas + BrandCountry
internal/asn/orgregex_test.go
internal/asn/mmdb.go              # dual-format MaxMind + ipinfo
internal/classifier/classifier.go # +RDAPLookup iface, +RegistrantCountry, +ASNCount
internal/classifier/decision.go   # refactor → Signals struct, voting, new tiers
internal/classifier/classifier_test.go
internal/rdap/bootstrap.go        # IANA embed + parse
internal/rdap/client.go           # HTTP + related-link follow
internal/rdap/cache.go            # disk cache 7d TTL
internal/rdap/parse.go            # vcardArray extractor
internal/rdap/rdap_test.go
internal/rdap/dns.json            # IANA bootstrap snapshot
internal/output/output.go         # +asn_country, +registrant_country
pkg/regionchecker/client.go       # Result mirror fields
cmd/regionchecker/subcommands.go  # --rdap, --rdap-timeout, --all, autoMMDB()
README.md
tasks/todo.md
```